AAKASH T C S | AI ENGINEER

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TECHNICAL SKILLS

Programming: Python, SQL
LLM & Agent Frameworks: LangChain, LangGraph, Model Context Protocol (MCP), ReAct agents, prompt engineering, tool calling, stateful workflows
RAG & Retrieval: Retrieval-Augmented Generation (RAG), GraphRAG, contextual retrieval, semantic chunking, embeddings, reranking, vector search, grounded generation
Models & Platforms: Azure OpenAI (GPT-4o), Google Gemini / Vertex AI, Amazon Bedrock (Claude), Groq
Data & Vector Stores: Pandas, MySQL, BigQuery, ArangoDB, Neo4j, FAISS, Chroma, Databricks Vector Search
Backend & Interfaces: REST APIs, Flask, FastAPI, Streamlit, Server-Sent Events (SSE)
Cloud & Serverless: Microsoft Azure, Google Cloud, AWS Lambda, Azure Functions

WORK EXPERIENCE

Movate | AI Engineer Jun 2024 – Present

- Designed and refined prompts and RAG workflows for enterprise LLM applications, using contextual enrichment to ground generated responses.
- Built modular LangChain/LangGraph workflows with conditional routing, tool invocation, and conversation state for multi-step tasks.
- Developed reusable Python modules for Dialogflow intent-creation automation, web scraping, and data preprocessing across translation and business-process workflows.
- Integrated Azure OpenAI, Gemini/Vertex AI, Amazon Bedrock (Claude), and Groq with Python services and REST APIs.
- Implemented Flask/REST integrations and server-sent events (SSE) streaming for conversational responses; applied grounding metadata and model safety settings.
- Contributed to enterprise prototypes involving agentic AI, RAG-to-DAX automation, customer-support analysis, fraud analysis, and semi-agentic chatbots.
- **Client engagements: Alcatel-Lucent-Enterprises and Nokia.**

PROJECTS

Agent-Native Conversational Chatbot | ALE | Python • LangGraph • Amazon Bedrock (Claude) • Neo4j • MCP • SSE

- Designed a **LangGraph orchestrator to route user intents to specialized agents for enterprise support workflows.**
- Built a **Neo4j knowledge graph for intent identification**, using Cypher-based GraphRAG retrieval to map user queries to the right agent.
- Exposed **Salesforce APIs as MCP-based tools** for a ReAct-style agent to create support tickets when requested by users.
- Added **logging for observability, safety guardrails, and checkpoint-based conversation history.**
- Implemented **thread/session separation for user-specific conversation state and privacy.**
- Used by 200+ users in production on a regular basis.

AI-Powered Document Extraction & Analytics View | Nokia | Python • LangGraph • Private LLM Gateway • MySQL

- Built **Python** and **LLM workflows** to extract and structure information from business documents, storing results in MySQL.
- Orchestrated **LangGraph workflows to route natural-language chatbot requests and analytics questions to a SQL-based agent**.
- Integrated an enterprise **Private LLM Gateway for document understanding and information extraction.**
- Exposed structured outputs for downstream reporting, analytics, and information retrieval.
- Used by Network engineers for technical query assistance and analytics view.

Natural-Language-to-Pandas Analytics App | Python • Streamlit • Pandas • Groq • Gemini

- Built an application that converts natural-language questions and DataFrame schemas into generated pandas filtering logic for business datasets.
- Added syntax and logic checks, generated test-case validation, and bounded retries to catch invalid or incorrect functions.

Grounded Q&A with Vertex AI Vector Search | Python • Vertex AI • Gemini • BGE Embeddings

- Created a Vertex AI Vector Search index and endpoint workflow using BAAI/bge-small-en-v1.5 embeddings for semantic retrieval.
- Connected retrieved context to Gemini-based answer generation with attention to grounding, retrieval quality, and safety settings.

EDUCATION

Velammal Engineering College, Chennai 2020 – 2024
Bachelor of Engineering – Computer Science and Engineering
CGPA: 8.4 / 10

CERTIFICATIONS & ADDITIONAL EXPERIENCE

- Google Data Analytics Certificate — Coursera
- C Programming Course
- Web Development Intern / Trainee — NSIC Training Centre (Jan 2023)
- Java Intern — Shiash Infotech Solutions (Aug–Oct 2022)
- Technical blog contribution on low-code GenAI business-process automation; judge at a 24-hour Velammal hackathon.